

- 6 W is a long straight wire of length 1.2 m. A beam of electrons is projected with a speed of $2.1 \times 10^7 \text{ m s}^{-1}$ parallel to W at a distance of 20 cm from it, as shown in Fig. 6.1.

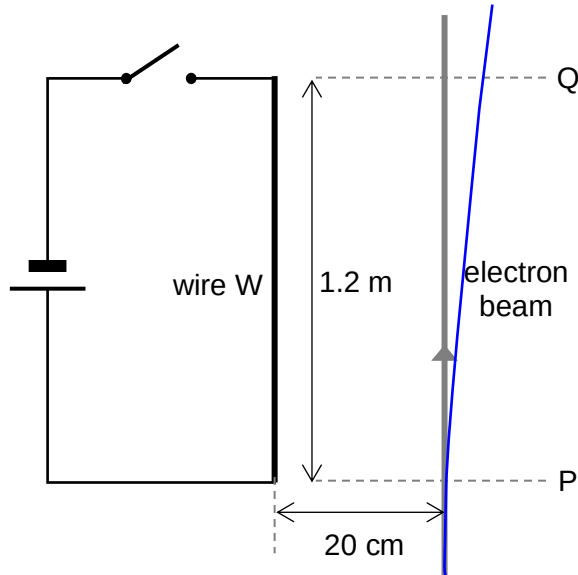


Fig. 6.1

- (a) When the switch is closed, the electron beam is deflected laterally by the current in W. Sketch the appearance of the deflected beam in Fig. 6.1. [1]
- (b) The current in W is 1.8 A. Calculate the flux density due to the current at a distance of 20 cm from W.

$$\begin{aligned}
 B &= \mu I / 2 \pi r \\
 &= 2 \times 10^{-7} \times 1.8 / 0.20 \quad [1] \\
 &= 1.8 \times 10^{-6} \text{ T} \quad [1]
 \end{aligned}$$

flux density = T [2]

- (c) Assuming that the lateral deflection of the electron beam is much less than 20 cm, show that the deflected path has a radius of 66 m.

$$F = m a$$

$$B q v = m v^2 / r \quad [1]$$

$$r = m v / B q$$

$$= 9.1 \times 10^{-31} \times 2.1 \times 10^7 / 1.8 \times 10^{-6} \times 1.6 \times 10^{-19} \quad [1]$$

[2]

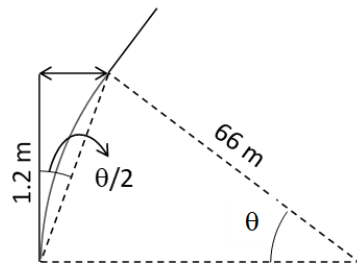
- (d) Calculate the lateral deflection of the beam at Q in mm.

$$\text{Angle traversed} = \theta = L / r$$

$$= 1.2 / 66 \quad [1] = 1.82 \times 10^{-2} \text{ rad}$$

$$\text{Deflection } x = L \tan(\theta/2) = 1.2 \times 0.91 \times 10^{-2} \quad [1]$$

$$= 1.1 \times 10^{-2} \text{ m} = 11 \text{ mm} \quad [1]$$



deflection = mm [3]

[Total: 8]

7. A 240 V a.c. mains is connected to a step-down transformer, a full wave rectifier and a motor as